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20231494

PRG521

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Source Code

```
using System;
using System.Collections.Generic;

// Interface for feedable animals public interface
IFeedable
{
   void Feed(string food);
}

// Interface for movable animals public interface
IMovable
```

```
{
   void Move();
}

// Base class for animals public class Animal
{
   // Common properties     public string Name { get;
set; }     public int Age { get; set; }

   // this is a constructor
   public Animal(string name, int age)
   {
      Name = name;
      Age = age;
   }

   public virtual void Eat(string food)
   {
      Console.WriteLine($"{Name} is eating {food}.");
   }

   // This is a common Method     public virtual void
Sleep()
   {
      Console.WriteLine($"{Name} is sleeping.");
   }

   public virtual void Speak()
   {
      Console.WriteLine($"{Name} makes a sound.");
   }
}

// these are the Animal classes public class Lion  :
Animal, IMovable
{
   public Lion (string name, int age) : base(name, age)
   {
   }

   // this overides the eat method     public override void
Eat(string food)
   {
      Console.WriteLine($"{Name} the Lion  is eating {food}.");
   }

   // this overides the move method     public void Move()
   {
      Console.WriteLine($"{Name} the Lion is running .");
   }
```

```
    // Override Speak method     public override
void Speak()
    {
        Console.WriteLine($"{Name} the Lion Roars.");
    }
}
public class Rhino : Animal, IMovable
{
    public Rhino(string name, int age) : base(name, age)
    {
    }

    // Overrides Eat method to accept different types of food     public override void Eat(string food)
    {
        Console.WriteLine($"{Name} the Rhino is eating {food}.");
    }
    public void Move()
    {
        Console.WriteLine($"{Name} the Rhino is Wallowing.");
    }
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Rhino Grunts.");
    }
}

public class Koala : Animal, IFeedable
{
    public Koala(string name, int age) : base(name, age)
    {
    }

    // Implement Feed method     public void
Feed(string food)
    {
        Console.WriteLine($"{Name} the Koala is eating {food}.");
    }
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Koala is Ballowing.");
    }
    public void Move()
    {
        Console.WriteLine($"{Name} the koala is Climbing.");
    }
}
class Program
{
    static List<Animal> animals = new List<Animal>();
```

```
  static void Main(string[] args)
  {
    Console.WriteLine("Welcome to your Virtual Zoo Management System!");

    bool exit = false;        while (!exit)
    {
      Console.WriteLine("\nPlease select an option:");
      Console.WriteLine("1. Please add an animal of your choice");
      Console.WriteLine("2. Interact with an animal");
      Console.WriteLine("3. Exit the Program!!!");
      Console.Write("Please Enter your choice: ");

      int choice;
      if (int.TryParse(Console.ReadLine(), out choice))
      {
        switch (choice)
        {                 case 1:
              AddAnAnimal();                    break;
case 2:
              InteractWithAnAnimal();                    break;
case 3:                   exit = true;                    break;
default:
              Console.WriteLine("The Choice is Invalid . Please try again.");
              break;
        }          }
else
      {
        Console.WriteLine("You have chosen wrong. Please enter a number.");
      }
    }
  }
  static void AddAnAnimal()
  {
    Console.WriteLine("\nChoose an animal of your liking to add:");
    Console.WriteLine("1. Lion ");
    Console.WriteLine("2. Rhino");
    Console.WriteLine("3. Koala");
    Console.Write("Enter your choice: ");

    int choice;
    if (int.TryParse(Console.ReadLine(), out choice))
    {
      Console.Write("Name of your Animal: ");          string name =
Console.ReadLine();          Console.Write("Age of the animal: ");          int
age;
      if (int.TryParse(Console.ReadLine(), out age))
      {
        switch (choice)
        {                 case 1:                    animals.Add(new Lion (name, age));
```

```
                Console.WriteLine("Lion has been added successfully!");                break;                case 2:
animals.Add(new Rhino(name, age));
                Console.WriteLine("Rhinohas been added successfully!");                break;                case 3:
animals.Add(new Koala(name, age));
                Console.WriteLine("Koala has been added successfully!");                break;                default:
                Console.WriteLine("An Invalid choice has been made.");                break;
            }           }
else
        {
            Console.WriteLine("Age is Invalid. Please enter a number.");
        }       }
else
      {
        Console.WriteLine("Input is invalid. Please enter a number.");
      }
   }
   static void InteractWithAnAnimal()     {
      if (animals.Count == 0)
      {
        Console.WriteLine("There aren't any animals in the zoo.");            return;
      }

      Console.WriteLine("\nSelect an animal to interact with:");        for (int i = 0; i < animals.Count;
i++)
      {
        Console.WriteLine($"{i + 1}. {animals[i].Name}");        }
      Console.Write("Enter the number of the animal: ");

      int choice;
      if (int.TryParse(Console.ReadLine(), out choice) && choice >= 1 && choice <= animals.Count)
      {
        Animal selectedAnimal = animals[choice - 1];

        Console.WriteLine($"\nInteracting with {selectedAnimal.Name}:");
        Console.WriteLine("1. Feed");
        Console.WriteLine("2. Sleep");
Console.WriteLine("3. Speak");            if (selectedAnimal is
IMovable)
        {
            Console.WriteLine("4. Move");
        }

        Console.Write("Enter your choice: ");            int action;
        if (int.TryParse(Console.ReadLine(), out action))
        {
            switch (action)
            {                case 1:                if (selectedAnimal is IFeedable)
                {
                    Console.Write("Enter the type of food: ");                string food = Console.ReadLine();
(selectedAnimal as IFeedable).Feed(food);
```

```
                }                    else
                {
                    Console.WriteLine("This animal cannot be fed.");
                }                    break;                    case 2:
selectedAnimal.Sleep();                    break;                    case 3:
selectedAnimal.Speak();                    break;                    case 4:                    if
(selectedAnimal is IMovable)
                {
                    (selectedAnimal as IMovable).Move();
                }                    else
{
                    Console.WriteLine("This animal cannot move.");
                }                    break;
default:
                Console.WriteLine("Invalid choice.");                    break;
            }           }
else
        {
            Console.WriteLine("Invalid input. Please enter a number.");
        }        }
else
      {
        Console.WriteLine("Invalid input. Please enter a valid number.");
      }
   } }
```

Screenshots of source code

```
using System;
using System.Collections.Generic;

// Interface for feedable animals
3 references
public interface IFeedable
{
    2 references
    void Feed(string food);
}

// Interface for movable animals
5 references
public interface IMovable
{
    3 references
    void Move();
}

// Base class for animals
10 references
public class Animal
{
    // Common properties
    15 references
    public string Name { get; set; }
    1 reference
    public int Age { get; set; }

    // this is a constructor
    3 references
```

```
    public Animal(string name, int age)
    {
        Name = name;
        Age = age;
    }


    2 references
    public virtual void Eat(string food)
    {
        Console.WriteLine($"{Name} is eating {food}.");
    }

    // This is a common Method
    1 reference
    public virtual void Sleep()
    {
        Console.WriteLine($"{Name} is sleeping.");
    }


    4 references
    public virtual void Speak()
    {
        Console.WriteLine($"{Name} makes a sound.");
    }
}
string Animal.Name { get; set; }

// these are the Animal classes
2 references
public class Lion  : Animal, IMovable
```

```
{

    1 reference
    public Lion (string name, int age) : base(name, age)
    {
    }

    // this overides the eat method
    1 reference
    public override void Eat(string food)
    {
        Console.WriteLine($"{Name} the Lion  is eating {food}.");
    }

    // this overides the move method
    2 references
    public void Move()
    {
        Console.WriteLine($"{Name} the Lion is running .");
    }

    // Override Speak method
    2 references
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Lion Roars.");
    }
}

2 references
```

```
public class Rhino : Animal, IMovable
{
    1 reference
    public Rhino(string name, int age) : base(name, age)
    {
    }

    // Overrides Eat method to accept different types of food
    1 reference
    public override void Eat(string food)
    {
        Console.WriteLine($"{Name} the Rhino is eating {food}.");
    }

    2 references
    public void Move()
    {
        Console.WriteLine($"{Name} the Rhino is Wallowing.");
    }

    2 references
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Rhino Grunts.");
    }
}

2 references
public class Koala : Animal, IFeedable
{
    1 reference
```

```
    public Koala(string name, int age) : base(name, age)
    {
    }

    // Implement Feed method
    2 references
    public void Feed(string food)
    {
        Console.WriteLine($"{Name} the Koala is eating {food}.");
    }

    2 references
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Koala is Ballowing.");
    }
    0 references
    public void Move()
    {
        Console.WriteLine($"{Name} the koala is Climbing.");
    }
}

0 references
class Program
{
    static List<Animal> animals = new List<Animal>();

    0 references
    static void Main(string[] args)
    {
```

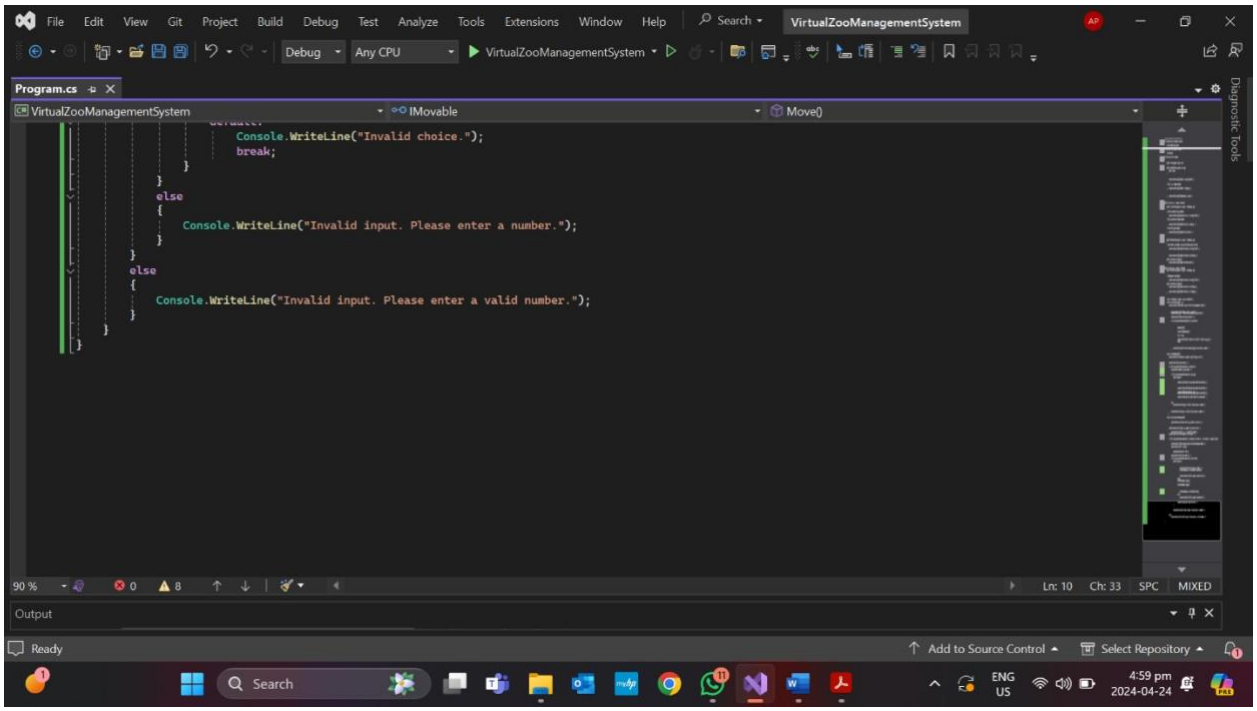
```
    {
        Console.WriteLine("Welcome to your Virtual Zoo Management System!");

        bool exit = false;
        while (!exit)
        {
            Console.WriteLine("\nPlease select an option:");
            Console.WriteLine("1. Please add an animal of your choice");
            Console.WriteLine("2. Interact with an animal");
            Console.WriteLine("3. Exit the Program!!!");
            Console.Write("Please Enter your choice: ");

            int choice;
            if (int.TryParse(Console.ReadLine(), out choice))
            {
                switch (choice)
                {
                    case 1:
                        AddAnAnimal();
                        break;
                    case 2:
                        InteractWithAnAnimal();
                        break;
                    case 3:
                        exit = true;
                        break;
                    default:
                        Console.WriteLine("The Choice is Invalid . Please try again.");
                        break;
                }
            }
```

```
            }
            else
            {
                Console.WriteLine("You have chosen wrong. Please enter a number.");
            }
        }
    }

    1 reference
    static void AddAnAnimal()
    {
        Console.WriteLine("\nChoose an animal of your liking to add:");
        Console.WriteLine("1. Lion ");
        Console.WriteLine("2. Rhino");
        Console.WriteLine("3. Koala");
        Console.Write("Enter your choice: ");

        int choice;
        if (int.TryParse(Console.ReadLine(), out choice))
        {
            Console.Write("Name of your Animal: ");
            string name = Console.ReadLine();
            Console.Write("Age of the animal: ");
            int age;
            if (int.TryParse(Console.ReadLine(), out age))
            {
                switch (choice)
                {
                    case 1:
                        animals.Add(new Lion (name, age));
                        Console.WriteLine("Lion has been added successfully!");
```

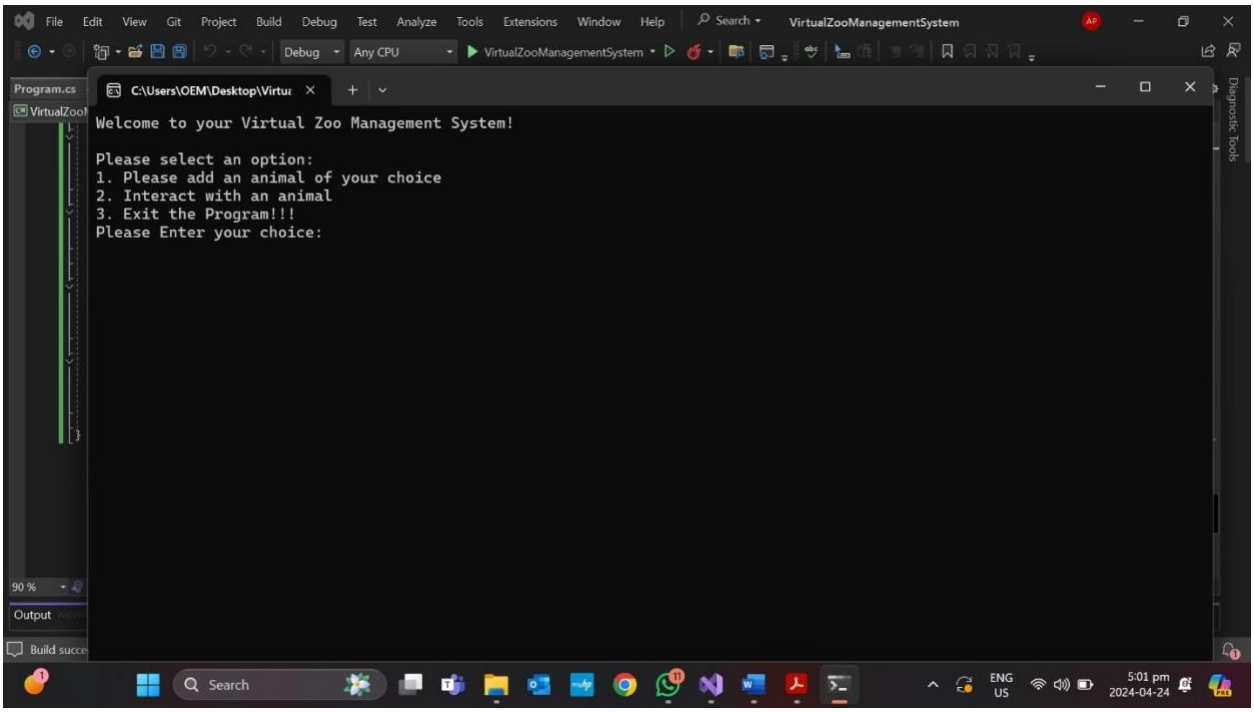
```
                        Console.WriteLine("Lion has been added successfully!");
                        break;
                    case 2:
                        animals.Add(new Rhino(name, age));
                        Console.WriteLine("Rhinohas been added successfully!");
                        break;
                    case 3:
                        animals.Add(new Koala(name, age));
                        Console.WriteLine("Koala has been added successfully!");
                        break;
                    default:
                        Console.WriteLine("An Invalid choice has been made.");
                        break;
                }
            }
            else
            {
                Console.WriteLine("Age is Invalid. Please enter a number.");
            }
        }
        else
        {
            Console.WriteLine("Input is invalid. Please enter a number.");
        }
    }

    1 reference
    static void InteractWithAnAnimal()
    {
        if (animals.Count == 0)
        {
```

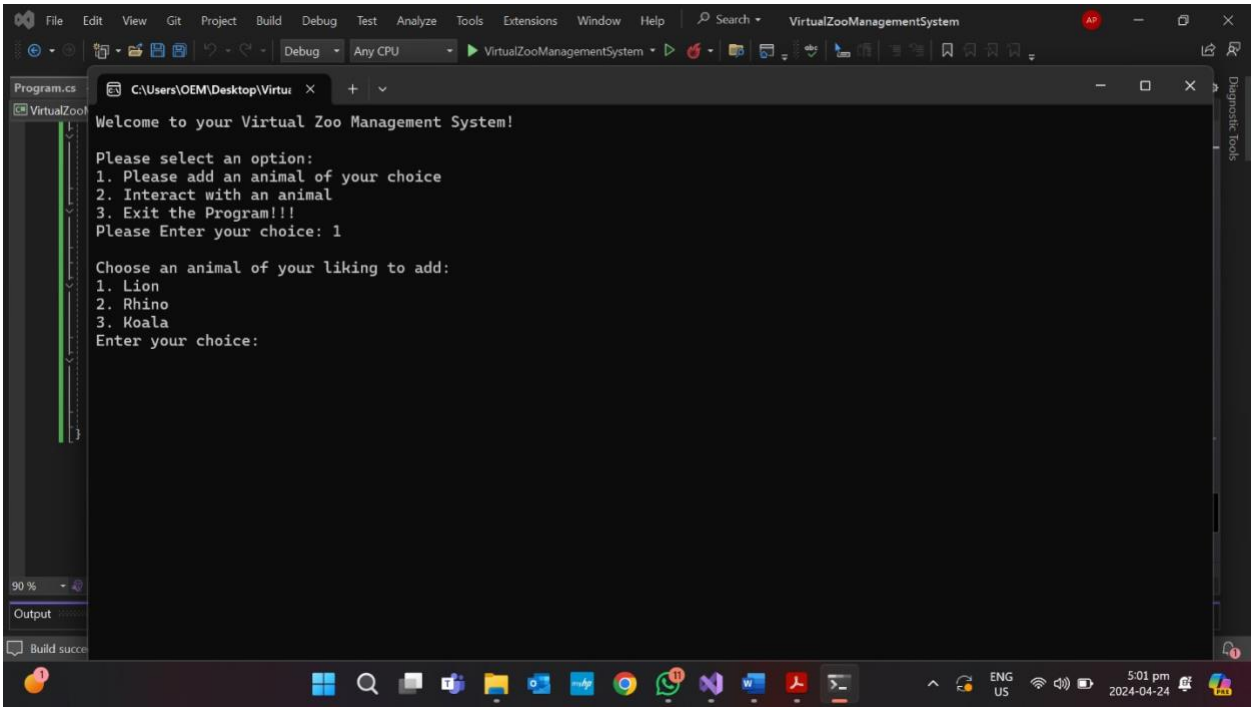
```
            Console.WriteLine("There aren't any animals in the zoo.");
            return;
        }

        Console.WriteLine("\nSelect an animal to interact with:");
        for (int i = 0; i < animals.Count; i++)
        {
            Console.WriteLine($"{i + 1}. {animals[i].Name}");
        }
        Console.Write("Enter the number of the animal: ");

        int choice;
        if (int.TryParse(Console.ReadLine(), out choice) && choice >= 1 && choice <= animals.Count)
        {
            Animal selectedAnimal = animals[choice - 1];

            Console.WriteLine($"\nInteracting with {selectedAnimal.Name}:");
            Console.WriteLine("1. Feed");
            Console.WriteLine("2. Sleep");
            Console.WriteLine("3. Speak");
            if (selectedAnimal is IMovable)
            {
                Console.WriteLine("4. Move");
            }

            Console.Write("Enter your choice: ");
            int action;
            if (int.TryParse(Console.ReadLine(), out action))
            {
                switch (action)
```

```
                {
                    case 1:
                        if (selectedAnimal is IFeedable)
                        {
                            Console.Write("Enter the type of food: ");
                            string food = Console.ReadLine();
                            (selectedAnimal as IFeedable).Feed(food);
                        }
                        else
                        {
                            Console.WriteLine("This animal cannot be fed.");
                        }
                        break;
                    case 2:
                        selectedAnimal.Sleep();
                        break;
                    case 3:
                        selectedAnimal.Speak();
                        break;
                    case 4:
                        if (selectedAnimal is IMovable)
                        {
                            (selectedAnimal as IMovable).Move();
                        }
                        else
                        {
                            Console.WriteLine("This animal cannot move.");
                        }
                        break;
                    default:
```


File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search
VirtualZooManagementSystem
Debug
Any CPU
VirtualZooManagementSystem
Program.cs
VirtualZooManagementSystem
IMovable
Move()
Console.WriteLine("Invalid choice.");
break;
}
}
else
{
Console.WriteLine("Invalid input. Please enter a number.");
}
}
else
{
Console.WriteLine("Invalid input. Please enter a valid number.");
}
}
}
90 %
0
8
Ln: 10
Ch: 33
SPC
MIXED
Output
Ready
Add to Source Control
Select Repository
Search
ENG
US
4:59 pm
2024-04-24

Screenshots of Output





```
Welcome to your Virtual Zoo Management System!

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 1

Choose an animal of your liking to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Name of your Animal: Simba
Age of the animal: 11
```

```
Welcome to your Virtual Zoo Management System!

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 1

Choose an animal of your liking to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Name of your Animal: Simba
Age of the animal: 11
Lion has been added successfully!

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 2

Select an animal to interact with:
1. Simba
Enter the number of the animal:
```

```
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 1
This animal cannot be fed.

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 2

Select an animal to interact with:
1. Simba
Enter the number of the animal: 1

Interacting with Simba:
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 2
Simba is sleeping.

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice:
```

```
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 2
Simba is sleeping.

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 2

Select an animal to interact with:
1. Simba
Enter the number of the animal: 1

Interacting with Simba:
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 3
Simba the Lion Roars.

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice:
```

```
C:\Users\OEM\Desktop\Virtu

1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 3
Simba the Lion Roars.

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 2

Select an animal to interact with:
1. Simba
Enter the number of the animal: 1

Interacting with Simba:
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 4
Simba the Lion is running .

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice:
```

```
C:\Users\OEM\Desktop\Virtu

2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 1

Choose an animal of your liking to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 2
Name of your Animal: Nathan
Age of the animal: 20
Rhinohas been added successfully!

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 2

Select an animal to interact with:
1. Simba
2. Nathan
Enter the number of the animal: 2

Interacting with Nathan:
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice:
```

```
Microsoft Visual Studio Debu

Rhinohas been added successfully!

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 2

Select an animal to interact with:
1. Simba
2. Nathan
Enter the number of the animal: 2

Interacting with Nathan:
1. Feed
2. Sleep
3. Speak
4. Move
Enter your choice: 1
This animal cannot be fed.

Please select an option:
1. Please add an animal of your choice
2. Interact with an animal
3. Exit the Program!!!
Please Enter your choice: 3

C:\Users\OEM\Desktop\VirtualZooManagementSystem\VirtualZooManagementSystem\bin\Debug\net8.0\VirtualZooManagementSystem.e
xe (process 8676) exited with code 0.
Press any key to close this window . . .
```

DECLARATION OF AUTHENTICITY

I Chane Marais hereby

(FULL NAME) declare that the contents of this assignment are entirely my own work except for the following elements: (List the elements of work in this project that were not self-generated as well as who the originator of the element is)

Signature: CM Date: 24-04-24